

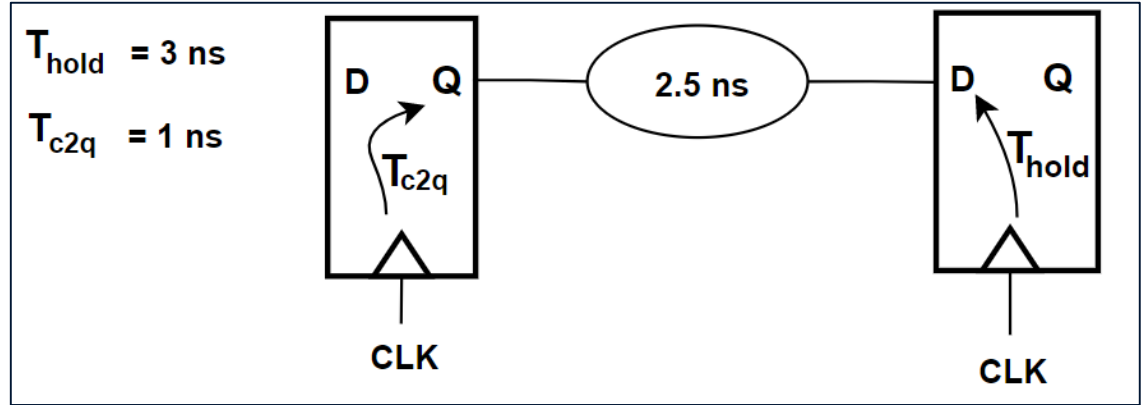
Example 1

- **Assumptions**

1. Interconnect Delays not considered
2. Clock Uncertainty not considered
3. One Operating Condition

- **Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack



$T_{path} = 1 + 2.5 = 3.5 \text{ ns}$
 $T_{hold} = 3 \text{ ns} < T_{path}$
Met Timing
 $T_{slack} = 3.5 - 3 = 0.5 \text{ +ve slack}$

Example 2

- Assumptions**

1. Interconnect Delays not considered
2. **Clock Uncertainty is considered**
3. One Operating Condition

- Required**

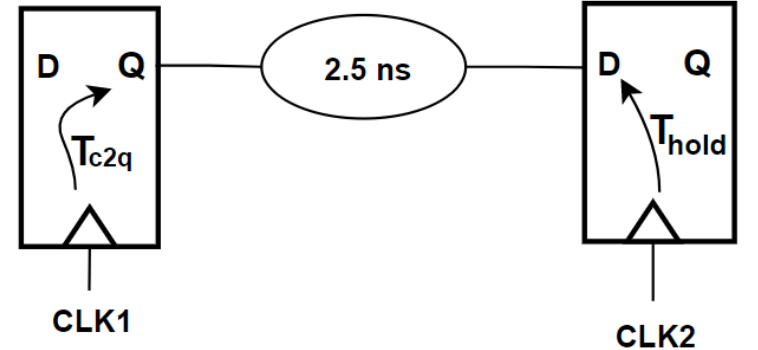
1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack

$$T_{\text{hold}} = 3 \text{ ns}$$

$$T_{\text{c2q}} = 1 \text{ ns}$$

Skew

$$\text{CLK 1_2} = 0.7 \text{ ns}$$



$$T_{\text{path}} = 2.5 + 1 = 3.5 \text{ ns}$$

$$T_{\text{hold}} + T_{\text{skew}} = 0.7 + 3 = 3.7 \text{ ns}$$

Violation

negative slack

$$T_{\text{slack}} = 3.5 - 3.7 = -0.2 \text{ ns}$$

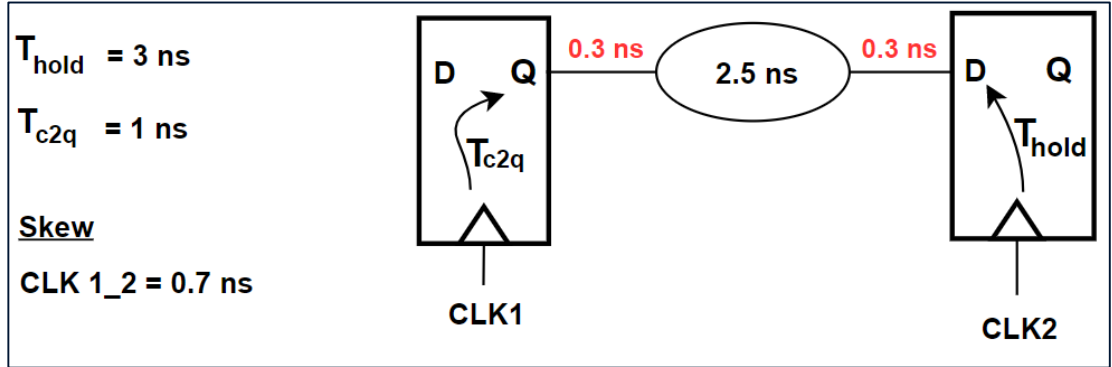
Example 3

- Assumptions**

1. Interconnect Delays is considered
2. Clock Uncertainty is considered
3. One Operating Condition

- Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack



$$T_{\text{path}} = 1 + 0.3 + 2.5 + 0.3 = 4.1 \text{ ns}$$
$$T_{\text{hold}} + T_{\text{skew}} = 3 + 0.7 = 3.7 \text{ ns}$$

No Violation
Positive Slack
 $T_{\text{slack}} = 4.1 - 3.7 = 0.4 \text{ ns}$

Example 4

$$T_{pd2} = 0.3 + 1.3 + 0.95 = 2.55\text{ns}$$

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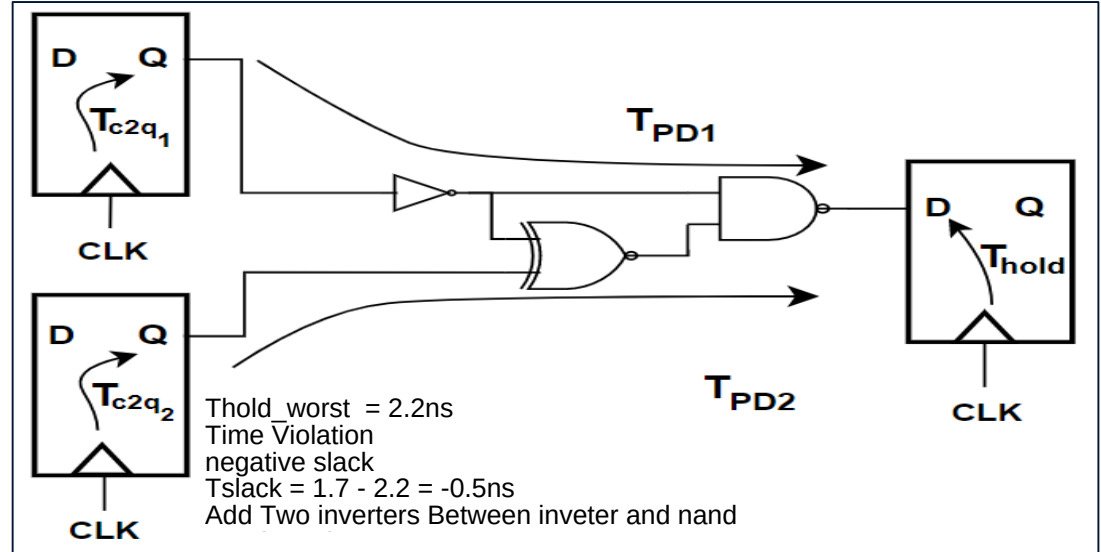
$$\text{Tpd3} = 0.3 + 0.45 + 1.3 + 0.95 = 3\text{ns}$$

- **Assumptions**

1. Interconnect Delays is not considered
2. Clock Uncertainty is not considered
3. **Worst/Best Operating Conditions**

- **Required**

1. Is this path timing is met or not?
2. if met calculate the positive slack
3. if not calculate the negative slack and suggest a solution to fix hold violation



Delays	Inverter	XNOR	NAND	Tcq1	Tcq2	Thold
Max	0.5	1.5	1	0.4	0.35	2.2
Min	0.45	1.3	0.95	0.3	0.3	2